

A finite-block positive case for the Kim–Lee direct-sum retraction question

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Source and question

The source is Sun Kwang Kim and Han Ju Lee, *Simultaneously continuous retraction and Bishop-Phelps-Bollobas type theorem*, arXiv:1308.1638. A dual space X^* is said to admit a uniformly simultaneously continuous retraction if there is a retraction $r : X^* \rightarrow B_{X^*}$ which is weak-star continuous and uniformly continuous for the norm topology.

After proving the ℓ_1 -sum stability theorem in Proposition 2.5, the paper records the following open point:

“We do not know that the similar result of [Proposition 2.5] holds for c_0 or ℓ_p sums for $1 < p < \infty$.”

Proposition 2.5. *Let $\{X_j\}_{j \in J}$ be a family of Banach spaces and let $X = [\bigoplus X_n]_1$. Suppose that each X_j^* admits a uniformly simultaneously continuous retraction φ_j . If*

$$\lim_{\varepsilon \rightarrow 0+} \sup_{j \in J} \omega_{\varphi_j}(\varepsilon) = 0,$$

then X^ admits a uniformly simultaneously continuous retraction. In particular, a finite ℓ_1 sum of Banach spaces whose duals admit uniformly simultaneously continuous retractions also admits a uniformly simultaneously continuous retraction.*

Proof. For each $x^* \in X^*$, define $\varphi(x^*) = (\varphi_j(x^*))_{j \in J}$. Then it is easy to check that φ is uniformly norm-continuous and weak-* continuous. $\square$

We do not know that the similar result of holds for c_0 or ℓ_p sums for $1 < p < \infty$. However, we provide a positive result for separable uniformly smooth spaces. Recall that a Banach space X is said to be uniformly convex if the modulus of convexity

$$\delta_X(\varepsilon) = \inf \left\{ 1 - \left\| \frac{x+y}{2} \right\| : x, y \in S_X, \text{ and } \|x-y\| \geq \varepsilon \right\}$$

is positive for all $0 < \varepsilon < 1$. A Banach space X is uniformly smooth if and only if X^* is uniformly convex. In the proof, we will use the following lemma.

The paper then gives a positive result for separable uniformly smooth summands whose duals have uniformly convex moduli bounded away from zero. The result below gives a different positive subcase: arbitrary finite-dimensional blocks, with no rotundity or smoothness assumption on their norms.

Idea of the proof

The usual radial projection $z \mapsto z/\max\{1, \|z\|\}$ is uniformly norm continuous, but it is not weak-star continuous on dual sequence spaces. The replacement is an ordered prefix truncation. Given $z = (z_n) \in (\oplus E_n)_{\ell_q}$, keep the initial blocks until the cumulative q -mass first reaches one, scale only that last block so that the new q -mass is exactly one, and erase the later tail. This is the same geometric idea as the standard truncation of ℓ_1 , but applied to block norms and q -powers.

The key point is that weak-star convergence sees every fixed finite group of blocks in norm when the blocks are finite-dimensional. If the crossing block does not stabilize, it can only run to infinity, and the erased or newly created tail disappears against the c_0 or ℓ_p predual. Norm continuity follows from a uniform Holder estimate for this first-crossing map.

The partial result

Theorem 1. *Let $(F_n)_{n \geq 1}$ be finite-dimensional real or complex Banach spaces. Let either*

$$X = \left(\bigoplus_{n=1}^{\infty} F_n \right)_{c_0} \quad \text{or} \quad X = \left(\bigoplus_{n=1}^{\infty} F_n \right)_{\ell_p}, \quad 1 < p < \infty.$$

Then X^ admits a uniformly simultaneously continuous retraction onto B_{X^*} .*

Consequently, in this finite-dimensional-block case, the source paper's direct-sum stability question has an affirmative answer for c_0 -sums and for ℓ_p -sums, $1 < p < \infty$.

Proof. Write

$$E_n = F_n^*, \quad Z = \left(\bigoplus_{n=1}^{\infty} E_n \right)_{\ell_q},$$

where $q = 1$ in the c_0 -sum case and $q = p/(p-1)$ in the ℓ_p -sum case. Then $Z = X^*$ with its canonical weak-star topology.

For $z = (z_n) \in Z$, put $a_n(z) = \|z_n\|$. If $\|z\|_q \leq 1$, define $Rz = z$. If $\|z\|_q > 1$, let $m = m(z)$ be the first integer such that

$$\sum_{n=1}^m a_n(z)^q \geq 1.$$

Then $\sum_{n < m} a_n(z)^q < 1$, and $a_m(z) > 0$. Define

$$\lambda(z) = \frac{(1 - \sum_{n < m} a_n(z)^q)^{1/q}}{a_m(z)} \in (0, 1],$$

and set

$$(Rz)_n = \begin{cases} z_n, & n < m, \\ \lambda(z)z_m, & n = m, \\ 0, & n > m. \end{cases}$$

It is immediate that $Rz \in B_Z$, that $\|Rz\|_q = 1$ whenever $\|z\|_q > 1$, and that R is the identity on B_Z . Thus R is a retraction onto B_Z .

We first record the norm estimate. For $1 \leq q < \infty$, the map R is uniformly continuous on every ℓ_q -sum of normed spaces. More precisely, there is a constant C_q such that

$$\|Ru - Rv\|_q \leq C_q \|u - v\|_q^{1/q} \quad \text{whenever } \|u - v\|_q \leq 1,$$

and the trivial bound $\|Ru - Rv\| \leq 2$ handles larger distances.

Here is the estimate. Let $\delta = \|u - v\|_q \leq 1$. If both u and v belong to B_Z , there is nothing to prove. If, say, $\|u\|_q \leq 1$ and $\|v\|_q > 1$, then $\|v\|_q \leq 1 + \delta$, and the erased part satisfies

$$\|v - Rv\|_q^q \leq \|v\|_q^q - 1 \leq (1 + \delta)^q - 1 \leq 2^q q \delta.$$

Hence $\|Ru - Rv\|_q \leq \delta + 2q^{1/q}\delta^{1/q}$.

It remains to consider the case where both points are outside B_Z . Let $m(u)$ and $m(v)$ be their crossing indices. If $m(u) = m(v) = m$, then all coordinates except m are either unchanged or erased in the same way. For the m -th coordinate, write $u_m = a\xi$, $v_m = b\eta$ with $\|\xi\| = \|\eta\| = 1$ when the coordinate is nonzero, and let

$$s = \left(1 - \sum_{n < m} \|u_n\|^q\right)^{1/q}, \quad t = \left(1 - \sum_{n < m} \|v_n\|^q\right)^{1/q}.$$

The scalar inequality $|\alpha^{1/q} - \beta^{1/q}| \leq |\alpha - \beta|^{1/q}$ for $\alpha, \beta \geq 0$, together with $\|P_{m-1}u - P_{m-1}v\|_q \leq \delta$, gives $|s - t| \leq C_q \delta^{1/q}$. Also $|a - b| \leq \|u_m - v_m\|$. Since

$$s\xi - t\eta = (u_m - v_m) - ((a - s)\xi - (b - t)\eta),$$

and

$$\|(a - s)\xi - (b - t)\eta\| \leq |(a - s) - (b - t)| + 2\|u_m - v_m\|,$$

we get $\|s\xi - t\eta\| \leq C_q \delta^{1/q}$. Adding the unchanged prefix gives the desired bound in the equal-crossing case.

If $m(u) < m(v)$, then the $m(u)$ -prefix of v has norm < 1 , while the $m(u)$ -prefix of u has norm at least 1. Since these prefixes are within δ , the $m(u)$ -prefix of u has norm at most $1 + \delta$. The same estimate as above shows that replacing this prefix by the truncated prefix Ru changes it by at most $C_q \delta^{1/q}$. Moreover, the part of Rv after the $m(u)$ -th block has q -mass at most

$$1 - \|P_{m(u)}v\|_q^q \leq C_q \delta.$$

Therefore $\|Ru - Rv\|_q \leq C_q \delta^{1/q}$. The case $m(v) < m(u)$ is symmetric. This proves uniform norm continuity.

It remains to prove weak-star continuity. Let $z_\alpha \rightarrow z$ weak-star in Z . Since the blocks F_n are finite-dimensional, each coordinate $(z_\alpha)_n \rightarrow z_n$ in norm, and hence every finite prefix norm converges. The net (Rz_α) is contained in B_Z , so coordinatewise weak-star convergence is enough to test against elements of the predual $(\oplus F_n)_{c_0}$ when $q = 1$, and against elements of $(\oplus F_n)_{\ell_p}$ when $1 < q < \infty$, by first truncating the predual vector to finitely many coordinates.

Fix a coordinate k . If z is outside the unit ball and its crossing index is m , then for every $j < m$ the crossing inequalities are eventually unchanged, so $(Rz_\alpha)_j \rightarrow (Rz)_j$. If the crossing of z is strict at m , the crossing index of z_α is eventually m , and the m -th coordinate also converges in norm. If the m -th prefix of z has norm exactly one, then later crossings of z_α , if they occur, have remaining q -mass tending to zero; hence every fixed coordinate after m tends to zero, while the first m coordinates converge to those of Rz .

If $z \in B_Z$, then each finite prefix of z has norm strictly less than one unless z has a finite prefix of norm one. In the first subcase the crossing index of z_α , when it exists, eventually lies after any prescribed finite prefix, so the fixed coordinates of Rz_α agree eventually with those of z_α . In the finite-prefix boundary subcase, the same zero-remaining-mass argument used above applies. Thus $(Rz_\alpha)_k \rightarrow (Rz)_k$ for each fixed k , and consequently $Rz_\alpha \rightarrow Rz$ weak-star.

Therefore $R : X^* \rightarrow B_{X^*}$ is a weak-star continuous, uniformly norm continuous retraction. $\square$

Verification and novelty notes

The construction was checked against the exact source passage on page 6 of arXiv:1308.1638. The run indexes were searched for ‘1308.1638’, ‘simultaneously continuous retraction’, ‘Bishop-Phelps-Bollobas’, ‘continuous retraction’, ‘BPBp’, and related direct-sum terms; no existing run packet for this paper or this finite-dimensional-block subcase was found. Web and citation checks found the source arXiv page and one Semantic Scholar citation, arXiv:1407.7872, concerning BPBp for operators from $C(K)$ into uniformly convex spaces rather than this retraction stability question. Local corpus searches for the exact retraction terminology found no later closure of the direct-sum stability problem.

The proof is independent of the later uniformly convex-dual theorem in the source paper. Its finite-dimensional hypothesis is essential for this method: it is used precisely to make coordinate norms weak-star continuous.

Human review recommendation

Recommended review focus: check the uniform-continuity estimate for the first-crossing truncation map, especially the equal-crossing case. The weak-star continuity argument is the conceptual core and should be compared with the source paper’s basis truncation proof.

References

- [1] S. K. Kim and H. J. Lee, *Simultaneously continuous retraction and Bishop-Phelps-Bollobas type theorem*, arXiv:1308.1638, 2013.